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Karl W. Muessig

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Paleomagnetic data on the basic igneous intrusions of the central Falcón Basin, Venezuela

Karl W. Muessig
Exxon Minerals Company
P.O. Box 2189
Houston, Texas 77001

ABSTRACT

Paleomagnetic data for basaltic intrusions in the central Falcón Basin document a secondary magnetization associated with alteration during Miocene time. The rocks are extensively altered; however, most of this alteration probably accompanied cooling soon after intrusion. Comparing these secondary magnetic signatures to those of other Cenozoic sites in South America, we conclude that magnetization occurred before the late Miocene folding in the basin and soon after intrusion.

A later stable partial overprinting of this secondary magnetization occurred in which the present geomagnetic field was superimposed on the stable Miocene remanence. At many sites it is difficult to remove this overprinting during demagnetization. The resultant magnetic signature has a steep positive inclination, which in the past has been difficult to interpret.

INTRODUCTION

The Falcón Basin is an extensional pullapart rift of Neogene age which formed within a complex right-lateral transcurrent fault system along the Venezuelan continental margin. The stratigraphic, structural, and tectonic history of the basin is documented in this volume (Muessig, 1984).

The main phase of basin subsidence and filling occurred during Oligocene and Miocene times. Alkaline to subalkaline basaltic sills and dikes intrude the Oligocene-Miocene sediments. One of these sills (La Guarabitas, Fig. 1) was dated at 22.9 ± 0.9 m.y. by K-Ar methods (Muessig, 1978). This age is the youngest basaltic activity reported in Venezuela. Only dacites from the Carúpano area in eastern Venezuela are younger (5 ± 0.5 m.y.; Santamaría and Schubert, 1974).

The Falcón Basin was later inverted by compressional events such that the central basin is now the deformed core of the Falcón Anticlinorium. However, there is little direct evidence for the timing of this deformation in the central basin due to total erosion of younger formations. Unconformities along the northern margin of the basin document some late Miocene folding (Zambrano and others, 1971).

A reconnaissance paleomagnetic study of two Falcón igneous sites was included in the work of Skerlec and Hargraves (1980). They found remanent magnetic vectors with southerly declinations and steep positive inclinations. Surrounding geologic structures do not support a post-intrusion folding event as a total explanation for the inclinations. Instead, a superimposition of two vectors was proposed: one primary reversed prefolding and a stable secondary normal postfolding (Skerlec and Hargraves, 1980).

This report details a more extensive paleomagnetic study of the intrusive rocks in the central basin. It was undertaken to further constrain the timing of magnetization with respect to the intrusive event and to deformation of the basin.

SAMPLING AND ANALYSIS

Samples for paleomagnetic analyses from a particular locality were collected from single continuous outcrops with distances between samples of 1 to 10 m. Cores (2.5 cm diameter) were obtained in the field with a portable drill or drilled from oriented samples in the laboratory. Field orientations were measured with both sun and magnetic compasses, and no significant directional differences were noted between these techniques. The remanent magnetism of cylindrical specimens (2.3 cm long) was measured

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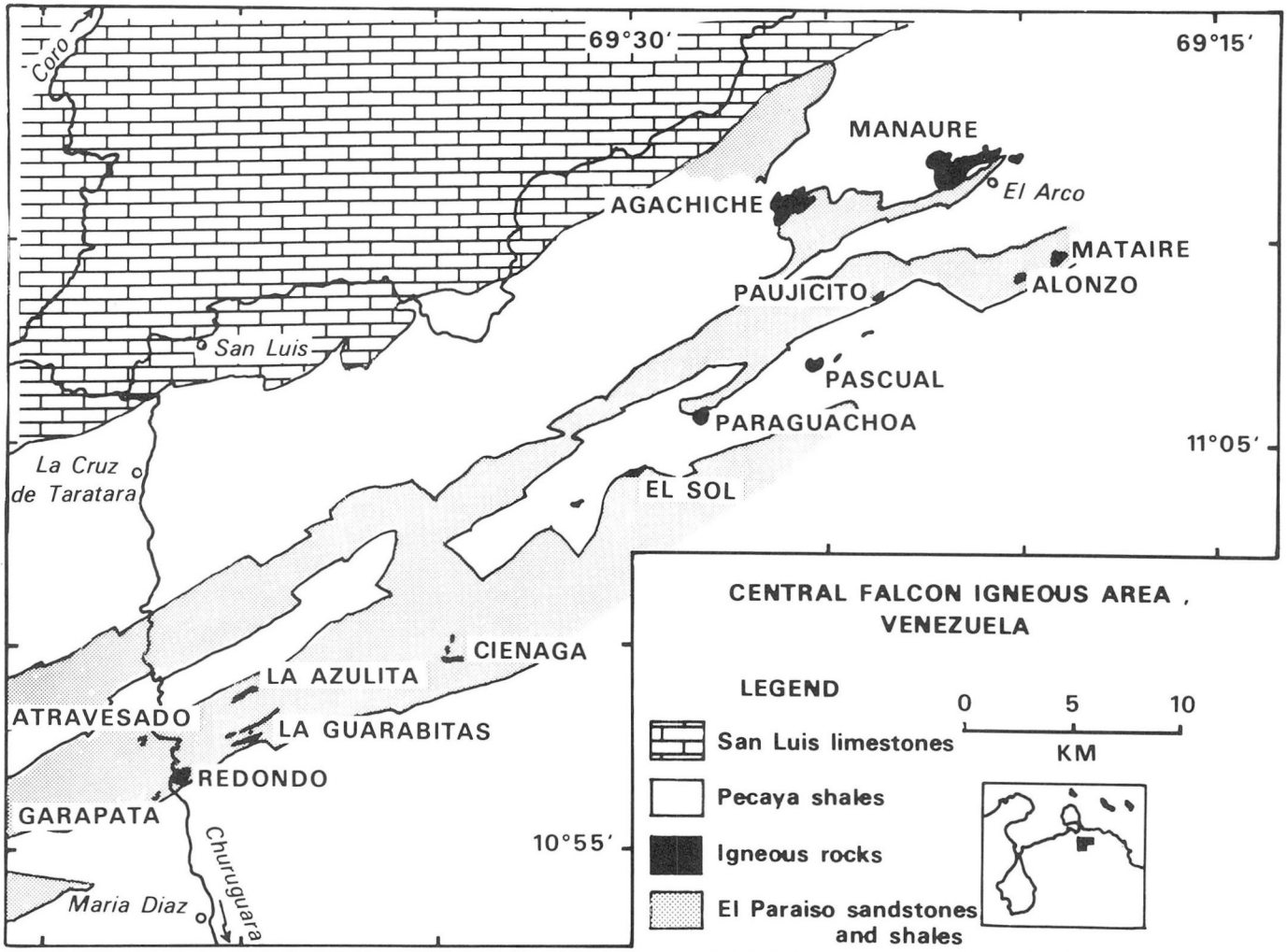


Figure 1. Location map of igneous intrusions in the central Falcón Basin. Paleomagnetic sample sites listed in Table 1 are identified by the name of the particular intrusion.

on a Princeton Applied Research SM-1 spinner magnetometer. The natural remanent magnetism (NRM) and the remanence after 50, 100, 150, 200, 300, and 500 Oe alternating field demagnetization were measured for each sample.

SAMPLE MINERALOGY

In all samples, with the exception of some from La Guarabitas, the primary basaltic mineralogy is extensively altered. Feldspars are replaced by saussurite, chlorite, albite, or sericite; biotite and amphibole when present are partially to completely chloritized; and iddingsite or chlorite are pseudomorphs after olivine. Fresh clinopyroxene is present in most samples; however, in other samples (Atravesado, Paraguachoa) it is marginally to completely altered to chlorite, carbonate, and sphene.

Oxide assemblages and textures indicate an advanced state of oxidation (Haggerty, 1976) and alteration. Pseudomorphic textures preserve oxidation-exsolution lamellae of ilmenite along {111} planes of original titanomagnetites. In many samples such textures are the only evidence of the prior existence of magnetite. Excepting Atravesado and La Guarabitas, all magnetite has been replaced by sphene or totally oxidized to assemblages of titanohematite, ilmenite, and rutile. Although oxy-exsolved ilmenite lamellae and primary ilmenite are more resistant to later alteration, similar oxidation and sphenitization products have evolved. Chlorite pseudomorphs of olivine and biotite occasionally contain fine-grained hematite and rutile+hematite, respectively. Chrome spinel xenocrysts are surrounded by titanomagnetite or its oxidized products.

The Atravesado samples contain a few large (up to 20-mm diameter) homogeneous titanomagnetites. These are rounded xenocrysts with rims of fine-grained, equant titanomagnetite. Some are composite grains containing magnetite, pyrrhotite, and pyrite. Fine ilmenite lenses have been oxy-exsolved from the magnetite margins. Aside from the large xenocrysts, all other Atravesado oxides are altered to hematite, rutile, and sphene.

The Guarabitas samples contain abundant, fine-grained, partially oxidized magnetite with a spongy texture. Ilmenite, an oxi-

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TABLE 1. PALEOMAGNETIC DATA FOR THE CENTRAL FALCON IGNEOUS ROCKS

Site	Location, deg			NRM							Demagnetized data				
	Lat,N	Long,W	n	D,deg	I,deg	k	α_{95} deg	J_N,emu/cm^3	X	Q	ODF Oe	D,deg	I,deg	k	α_{95} deg
Atravesado	10.96	69.71	7	313	30	9	24	6.9×10^{-6}	3.0×10^{-5}	0.24	200	27	56	1	90
Garapata	10.94	69.70	8	25	55	4	34	4.3×10^{-6}	3.7×10^{-5}	0.12	200	358	40	27	11
Redondo*	10.95	69.69	10	27	58	3	37	9.2×10^{-5}			300	282	26	4	26
Guarabitas	10.97	69.66	11	105	17	2	42	9.1×10^{-3}	1.2×10^{-3}	7.6	200	126	11	2	45
Cienaga 1*	10.99	69.58	6	353	81	3	45	2.6×10^{-6}			300	313	87	9	24
Cienaga 2	10.99	69.58	8	325	49	14	16	3.6×10^{-6}	4.7×10^{-5}	0.08	200	312	72	4	30
Paraguachoa	11.09	69.47	6	9	65	2	62	6.6×10^{-7}	2.4×10^{-5}	0.03	200	13	30	4	40
Agachiche	11.18	69.43	6	249	-51	2	74	1.8×10^{-6}	3.5×10^{-5}	0.05	200	192	-40	18	16
Manaure	11.19	69.37	11	345	28	8	17	3.0×10^{-6}	3.6×10^{-5}	0.08	200	1	16	54	6

Note: n is the number of samples; D is the eastward declination from north; I is the inclination, downward positive; k is Fisher's precision parameter; α_{95} is the semiangle of cone of 95% confidence; J_N is the intensity of remanent magnetization; X is the magnetic susceptibility; Q is the Koenigsberger ratio ($J_N/X{\cdot}H$); ODF is the optimum demagnetizing field. Underlined sites meet the statistical criteria $\alpha_{95} \pm 20.0^{\circ}$ after demagnetization. Cerros Atravesado, Redondo, and Cienaga are intrusive plugs; Cerros Garapata, Guarabitas, and Manaure are sills; and Paraguachoa is a tuffaceous agglomerate pile.

* Data from Skerlec, and Hargraves (1980).

dation product, and pyrite are present in minor amounts. Other oxide alteration products include a mineral identified as maghemite, which exhibits magnetic properties but has a high, hematite-like reflectivity; and an unidentified Fe, Ti silicate detected by electron microprobe analysis.

DATA

The very low intensities of remanent magnetism and low susceptibilities (Table 1) are consistent with the rather highly altered mineralogy. The NRM intensity is typically less than 10^{-5} emu/cm^3; Cerro Redondo ($<10^{-4}$ emu/cm^3) and La Guarabitas ($<10^{-2}$ emu/cm^3) are exceptions. The Q (Koenigsberger) ratios are also generally low (see Table 1).

At the 200 Oe demagnetization level, intensities have dropped to between the 10^{-6} to 10^{-7} emu/cm^3 level (except Guarabitas), but remanence directions of individual samples show little change from the 100- and 150-Oe levels. Demagnetization of most samples above 200 Oe causes an unstable and viscous remanent magnetism. Repeated demagnetization and measurement of individual specimens at the 500-Oe level frequently yields diametrically opposed directions. For most samples data for the 500-Oe demagnetization level are not reported, since they are generally unreliable. Even at the 200-Oe level, the individual sample directions for many sites are not well grouped.

For the site mean directions, an arbitrary statistical cut off of Fisherian $\alpha_{95} \leq 20^{\circ}$ has been chosen as a minimum criterion of reliability (after Hicken and others, 1972). Only Cerros Manaure, Garapata, and Agachiche meet this restriction; Figures 2A, 2B, and 2C plot their individual sample NRMs and remanent directions after 200-Oe demagnetization. These figures indicate that alternating field demagnetization greatly improves the clustering of data. The site means and α_{95} cones of confidence have been plotted with the site data.

Cerro Manaure exhibits the best data cluster of all collected sites. Remanent magnetism is stable at the 200-Oe as well as the 500-Oe levels. Fisherian statistics yield a northerly declination, a shallow positive inclination, and an α_{95} of 6°.

Cerro Garapata exhibits a northerly declination, a moderate positive inclination, and an α_{95} of 11°. Cerros Manaure and Garapata were magnetized during a normal polarity interval. Although Cerros Atravesado and Paraguachoa have α_{95}s, which do not meet the reliability criteria, they too have northerly declinations and positive inclinations.

Cerro Agachiche, on the other hand, displays a southerly declination and a moderate negative inclination with an α_{95} of 16°. This implies magnetization during a reversed polarity interval.

INTERPRETATION

From the previously described oxide assemblages and silicate oxidation-alteration phenomena, it is suggested that the measured remanence is held in secondary mineral products.

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Magnetite-ilmenite and ilmenite-hematite-rutile assemblages suggest initial oxidation at rather high temperatures (600 °C–700 °C; Haggerty, 1976). This was accompanied by oxidation and dueteric alteration of olivine. High temperature olivine oxidation is suggested by a few olivine pseudomorphs with associated secondary hematite (Haggerty and Baker, 1967). A later lower temperature alteration (200 °C–400 °C; Baker and Haggerty, 1967) may have completely overprinted the high temperature olivine alteration. It seems that the introduction of connate waters from the host rocks altered the feldspars and redistributed Ca into either calcite or prehnite and zeolites, depending on local conditions. Sphenitization of much of the magnetite probably occurred at this time. Some of the Fe was redistributed into chlorites. The minor amount of secondary oxides (hematite) produced from silicate oxidation, coupled with the destruction of most of the magnetite by oxidation and sphenitization, accounts for the present low magnetic intensities.

In light of the above evidence, the measured remanent directions were probably induced during oxidation and alteration of the Falcón igneous bodies soon after intrusion. If intrusion, cooling, and initial alteration all occurred within a short time interval, then the primary and proposed secondary magnetization would be virtually identical in direction.

Considering that sites give approximately reversed northerly and southerly directions, it is fortuitous to assume that intrusion, cooling, and alteration occurred elsewhere with later allochthonous emplacement *without rotation* into the central Falcón area. Indeed, the directions of remanent magnetism in the Falcón rocks are distinctly different from the easterly directions observed for rocks of inferred allochthonous nature in northern Venezuela and the offshore islands (Skerlec and Hargraves, 1980). Paleomagnetic data for the Falcón rocks are consistent with an autochthonous intrusive origin.

Constraints on the timing of post-intrusion magnetization can be obtained from a study of the relationship of magnetization to folding in the Falcón Basin. In the central Falcón area the igneous intrusions are situated in strata that are folded around east- to east-northeast-trending axes. The Garapata, Agachiche, and Manaure sites are all conformable sills. If magnetization occurred prior to folding, then the remanent directions should be corrected to bring the folded strata back to a horizontal position. Because of the orientations of fold axes, the structural corrections (Table 2) have little effect on declination but substantially decrease inclination (Fig. 2D).

TABLE 2. VIRTUAL GEOMAGNETIC POLE (VGP) FOR FALCON SITES WITH $\alpha 95 \leq 20^{\circ}$, at 200-0e DEMAGNETIZATION BEFORE AND AFTER STRUCTURAL CORRECTIONS

	200 0e		VGP		Structurally corrected				VGP	
	D	I	Lat,N	Long,E	DIR	DIP	D	I	Lat,N	Long,E
Garapata	358	40	78	282	0	15	358	25	87	255
Agachiche	192	-40	73	334	350	41	186	-1	77	78
Manaure	1	16	87	90	355	15	1	1	80	106
Mean	5	33	(k=32,α_{95}=22)				3	8	(k=28,α_{95}=24)	
Mean VGP for 3 sites:			82	317					83	93
			(K=57,A_{95}=17)						(K=82,A_{95}=13)	

Note: D is the eastward declination from north; I is the inclination downward. For the applied structural correction DIR is the direction of dip eastward from north; DIP is the amount of dip from the horizontal. VGP for Agachiche has been reversed in calculating Mean VGP. All data are in units of degrees.

Because the bedding attitude is similar for the three sites, Fisher's k changes little with structural corrections (k=32 in situ versus k=28 structurally corrected; Table 2).

The validity of the structural corrections relies upon comparison of the mean VGP from the Falcón sites with other published Tertiary and Late Cretaceous poles for South America (Table 3). Although rather sparse, the available data for this time period all plot within a narrow, elongate swath (Fig. 3). The structurally corrected, mean VGP for Falcón plots within this swath, whereas the uncorrected mean VGP does not. In addition, the corrected Falcón VGP fits nicely between the group of Eocene and older VGP and the group of Pliocene and younger VGP. This would date the magnetization at Oligocene-Miocene, and thus it is consistent with the age of intrusion and alteration.

TABLE 3. CENOZOIC AND LATE CRETACEOUS PALEOMAGNETIC DATA FOR SOUTH AMERICA

Site	Location, deg		Age	D,deg	I,deg	VGP,deg		Reference
	Lat,N	Long,W	m.y.			Lat,N	Long,E	
Cerro del Fraile, Arg.	-50.0	72.7	1.0-2.5	352	-65	83	226	Fleck et al., 1972
Basalts, W. Arg.	-37.0	70.0	0.2-0.7	358	-60	85	131	Career and Valencio, 1970
Basalts, W. Arg.	-37.0	70.0	0.2-8.6	359	-59	87	132	"
Basalts, W. Arg.	-37.0	70.0	1.8-2.6	4	-60	85	75	"
Basalts, W. Arg.	-37.0	70.0	5.4-8.6	356	-57	86	194	"
Falcón, Venezuela*	11.1	69.5	23-28			83	93	This study
Abrolho Is., Braz.	-18.0	39.0	46-52	213	31	59	54	Pacca & Valencio, 1972
Rio de los Molino, Arg.	-31.8	64.5	63-68	165	53	77	198	Linares & Valencio, 1975
Pocos de Caldas, Braz.	-21.9	46.6	75	190	41	81	53	Opdyke & MacDonald, 1973

Note: For site locations, N latitude is positive, whereas S is negative.
*The listed age for the Falcón site is the possible range of intrusion ages (Muessig, 1978). The age of magnetization may be younger, but probably no younger than lower Miocene. The VGP is for the structurally corrected Falcón data.

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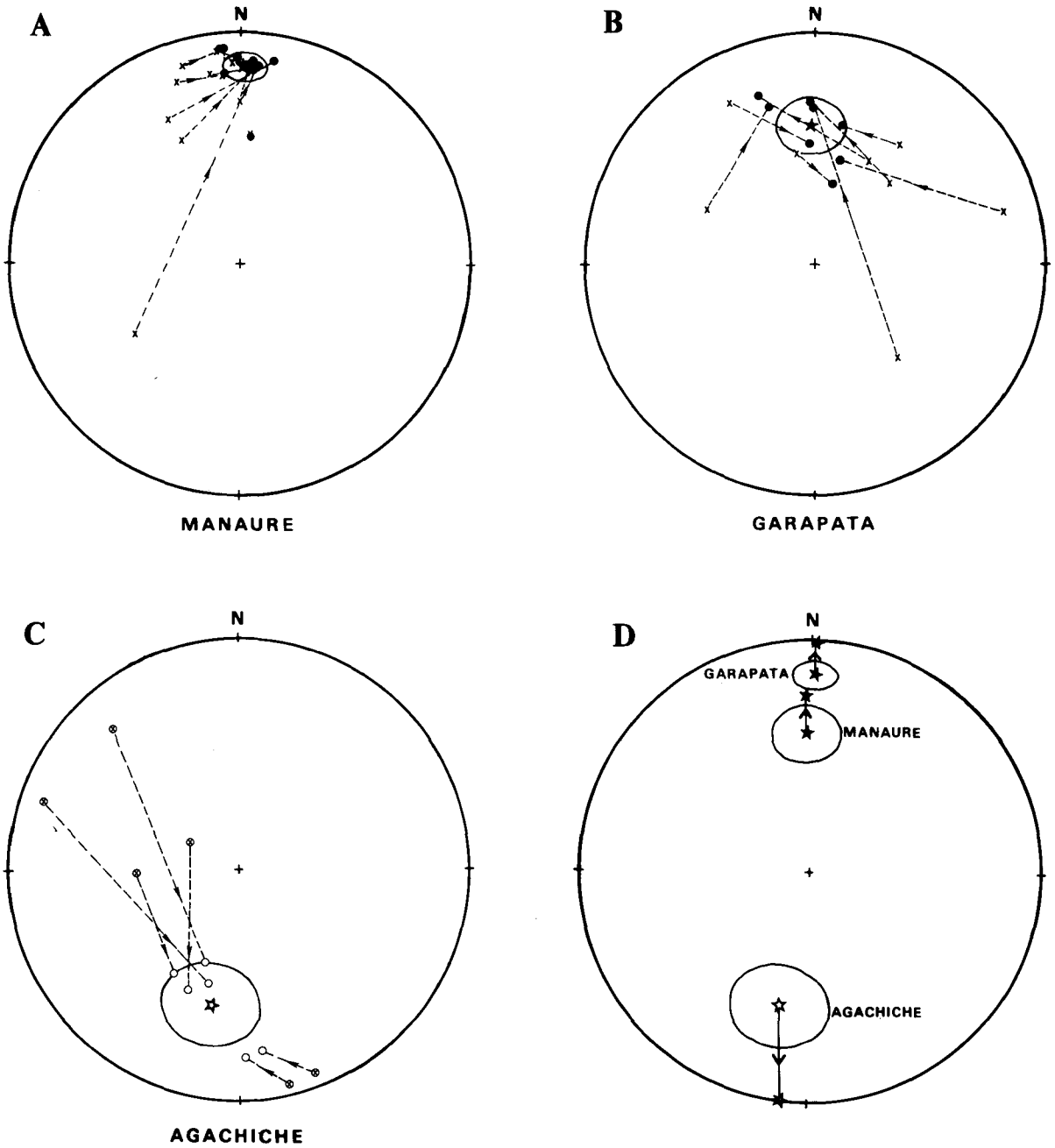


Figure 2. Equal area stereographic plots of directions of magnetization for Falcón paleomagnetic sites with $\alpha 95 < 20$. For NRM: x is a positive inclination; circled x is negative. For 200 Oe demagnetization: filled circle is a positive inclination; open circle is negative. For the site means of demagnetized data: black star is a positive inclination; open star is negative. Elipses indicate $\alpha 95$ confidences for site means of demagnetized data. For A, B, and C, arrows point toward demagnetized data. D is a plot of the directions of site means with and without structural corrections; arrows point toward structurally corrected data.

LATER REMAGNETIZATION

There is good evidence that some of the intensity weakness and scatter in the data are due to an even later remanence overprinting the proposed stable secondary vector.

The paleomagnetic data for Cienaga 1, Redondo (Skerlec and Hargraves, 1980), and for Cienaga 2 have $\alpha_{95} > 20°$ at 200-Oe demagnetization (Table 1). For most samples the NRMs are northerly with steep positive inclinations. During demagnetization, remanent directions are moved progressively southward, while a northerly vector with about 403° inclination is erased (Fig. 4). The resultant site mean has a southerly declination with

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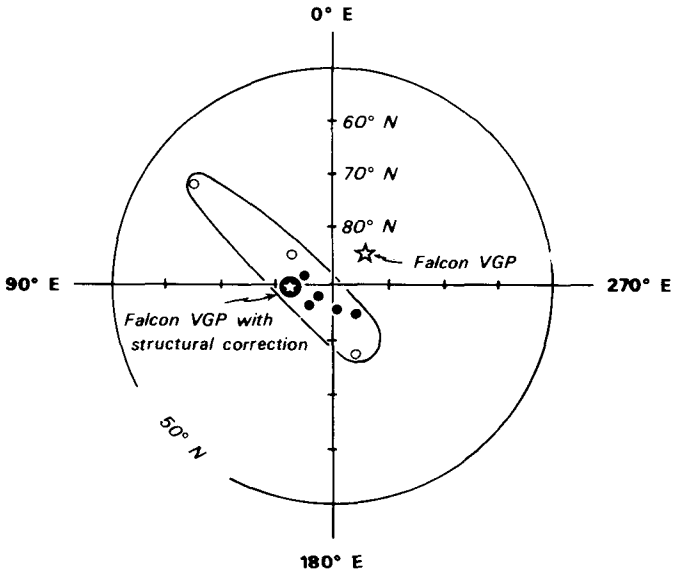


Figure 3. Equal area stereographic plot of Oligocene to Miocene virtual geomagnetic poles (VGP) for Falcón, Venezuela, and Cenozoic and Late Cretaceous poles for South America. Note that the structurally corrected VGP for Falcón falls within the field (elongate ellipse) of South American poles (pre-Oligocene open circle and post-Oligocene filled circle).

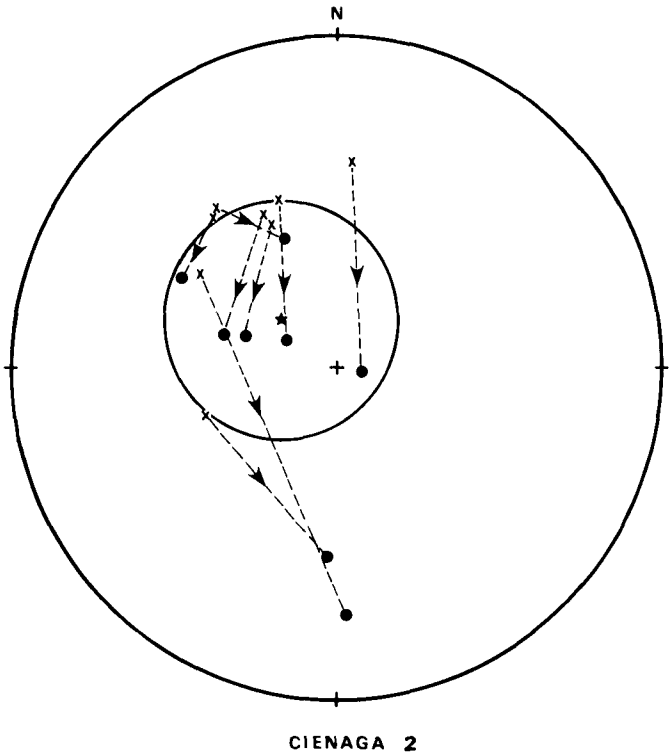


Figure 4. Equal area stereographic projection of directions of magnetization for Cerro Cienaga 2, Falcón. For NRM, x is a positive inclination. For 200 Oe demagnetization, filled circle is a positive inclination; black star is the site mean, and the ellipse is the α_{95} cone of confidence. Arrows connect the NRM with the 200 Oe data for each sample.

a steep positive inclination and a rather large scatter ($\alpha_{95} > 20°$). Not only is the steep inclination anomalous, but the positive nature would imply magnetization far south of the present position. A more feasible explanation depicts the remanent direction as a resultant of an original southern remanence with a nonparallel northerly overprint. The original remanence could be very similar to the structurally corrected Agachiche vector (D = 186.7 I = -1.0). The directions of the vectors erased during cleaning are consistent with the overprint being the present geomagnetic field.

Demagnetization to the 500-Oe level of samples from Redondo and Cienaga 1 (Skerlec and Hargraves, 1980) improves the statistics, moves the mean direction southward, but also fails to change the steep positive inclination. The data presented above support the proposal of Skerlec and Hargraves (1980) that the steep inclinations for some Falcón samples might be due to superimposition of secondary and tertiary magnetic vectors.

CONCLUSIONS

The above data suggest that the Falcón intrusions were probably magnetized after intrusion (22.9 m.y. ago) but prior to folding. Since folding in the Falcón Basin is thought to have occurred during late Miocene and Pliocene times (Zambrano and others, 1971), the age of magnetization is constrained to the early or middle Miocene. This stable remanence was most likely acquired soon after intrusion during the cooling and alteration of the rocks.

If so, then the consistent directions obtained from three sites indicate that the Falcón rocks are autochthonous intrusions. The steep inclinations of the remanent directions reported by previous investigators are best explained as a superposition of a very stable present geomagnetic field component upon the Miocene remanent vector.

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REFERENCES CITED

Baker, I., and Haggerty, S. E., 1967, The alteration of olivine in basaltic and associated lavas. Part II: Intermediate and low temperature alteration: Contributions to Mineralogy and Petrology, v. 16, p. 258–273.

Creer, K. M., and Valencio, D. A., 1970, Paleomagnetism and rock magnetic studies on the Cenozoic basalts from western Argentina: Geophysical Journal of the Royal Astronomical Society, v. 19, p. 113–146.

Fleck, R. J., Mercer, J. H., Nairn, A.E.M., and Peterson, D. N., 1972, Chronology of late Pliocene and early Pleistocene glacial and magnetic events in southern Argentina: Earth and Planetary Science Letters, v. 16, p. 15–22.

Haggerty, S. E., 1976, Oxidation of opaque mineral oxides in basalts, *in* Rumble, D., ed., Oxide minerals: Mineralogical Society of America Short Course Notes, v. 3, p. Hg1–Hg100.

Haggerty, S. E., and Baker, I., 1967, The alteration of olivine in basaltic and associated lavas. Part I: High temperature alteration: Contributions to Mineralogy and Petrology, v. 16, p. 233–257.

Hicken, A., Irving, E., Law, L. K., and Hastie, J., 1972, Catalogue of paleomagnetic directions and poles: Canadian Department of Energy, Mines and Resources, v. 45, p. 1–135.

Linares, E., and Valencio, D. A., 1975, Paleomagnetism and K-Ar ages of some trachybasaltic dikes from Rio de Los Molino, Province of Cordobo, Republic of Argentina: Journal of Geophysical Research, v. 80, p. 3315–3321.

Muessig, K. W., 1978, The central Falcón igneous suite, Venezuela: Alkaline basaltic intrusions of Oligocene-Miocene age: Geologie Mijnbouw, v. 57, p. 261–266.

Muessig, K. W., 1984, Structure and Cenozoic tectonics of the Falcón Basin, Venezuela, and adjacent areas, *in* Bonini, W. E., and others, eds., The Caribbean–South American plate boundary and regional tectonics: Geological Society of America Memoir 162 (this volume).

Opdyke, N. D., and MacDonald, W. D., 1973, Paleomagnetism of Pocos de Caldas Alkaline Complex, southern Brazil: Earth and Planetary Science Letters, v. 18, p. 37–44.

Pacca, I. G., and Valencio, D. A., 1972, Preliminary paleomagnetic study of igneous rocks from Abrolhos Islands, Brazil: Nature Physical Science, v. 240, p. 163–164.

Santamaría, F. J., and Schubert, C., 1974, Geochemistry and geochronology of the southern Caribbean–northern Venezuela plate boundary: Geological Society of America Bulletin, v. 85, p. 1085–1098.

Skerlec, G. M., and Hargraves, R. B., 1980, Tectonic significance of paleomagnetic data from northern Venezuela: Journal of Geophysical Research, v. 85, p. 5303–5315.

Zambrano, E., Vásquez, E., Duval, B., Latreille, M., and Coffinieres, B., 1971, Sintesis paleogeográfica y petrolera del occidente de Venezuela, *in* Memoria, Congreso Geológico Venezolano, IV, Caracas, 1969: Venezuela Ministerio de Minas e Hidrocarburos, Boletín de Geología, Publicación Especial no. 5, v. 1, p. 481–552.

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